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IMPROVED ELECTRIC BATTERY ASSEMBLY

Abstract

An electric battery assembly is disclosed comprising: a battery cell having an exterior housing and first and second electrical terminals enabling power to be drawn from the battery cell, the housing being electrically coupled to the first electrical terminal and electrically insulated from the second electrical terminal; an electronic unit comprising a measurement device; and wherein the electronic unit is electrically connected to the second electrical terminal and to the housing, thereby electrically coupling the electronic unit to the first battery cell terminal via the housing, enabling the measurement device to measure a property of the battery cell.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an electric battery assembly comprising a battery cell and an electronic unit, the electronic unit comprising a measurement device for measuring a property of the battery cell. In particular, means for electrically coupling the electronic unit to battery cell are disclosed to enable the measurement device to measure a property of the battery cell. In addition, battery packs comprising a plurality of the aforementioned battery assemblies are disclosed.

BACKGROUND

[0002] Known battery cells may comprise a measuring device arranged to measure a performance characteristic of the battery cell, such as the voltage or current output by the battery cell. Faulty and non-ideal operation of the battery cell may then be inferred from voltage, current, or other measurement data taken of the battery cell. Typically, this requires electrically connecting the measurement device to both electrical terminals of the battery cell. To achieve this, physical connections, for example using conductive wires, span the distance of separation of the terminals from the measuring device. For battery cells having small physical dimensions, this requirement does not present significant disadvantages during assembly of the battery. However, for physically larger form-factor batteries, for example, but not limited to, batteries where the distance of separation between the battery terminals is equal to or greater than 100 mm, having to physically connect the measuring device to both terminals via physical connections, such as wiring, increases the complexity and cost of assembly. Similar challenges are encountered where the cell terminals are located on different faces of the battery cell.

[0003] The routing of long wires between cell terminals and measurement device can present short circuit hazards. Where the measurement device is also powered by the battery cell it is electrically connected to, the use of long physical wires to provide the electrical connection may introduce errors in cell voltage measurements taken by the measurement device. This problem may be resolved by providing a Kelvin connection. However, this requires doubling the number of wires, and further increases the complexity of assembly.

[0004] It is an object of at least some embodiments of the present disclosure to address one or more shortcomings of the prior art, and in particular to provide a more convenient means for electrically coupling an electronic unit comprising a measurement device to a battery cell, which is also suitable for use with larger form-factor batteries and for batteries where the electrical terminals are located at different faces of the battery.

SUMMARY OF THE DISCLOSURE

[0005] In accordance with an aspect of the disclosure, there is provided an electric battery assembly comprising a battery cell and an electronic unit comprising a measurement device. The battery cell comprises an exterior housing and first and second electrical terminals enabling power to be drawn from the battery cell. The housing may be electrically coupled to the first electrical terminal and electrically insulated from the second electrical terminal. The electronic unit may be electrically connected to the second electrical terminal and to the housing, thereby electrically coupling the electronic unit to the first battery cell terminal via the housing, enabling the measurement device to measure a property of the battery cell. In this way, an electrical connection may be established

between the electrical terminals of the battery cell and the electronic device without requiring the use of electrical wiring spanning the distance of separation between the cell terminals and the electronic unit. The electronic unit may also be electrically powered by the battery cell. This is advantageous when the electronic unit is not provided with an alternative power source. The first and second electrical terminals may relate to respectively the positive and negative electrical terminals of the battery cell, or to the negative and positive electrical terminals of the battery cell.

[0006] The electric battery assembly may comprise an electrical insulator configured to insulate the second electrical terminal from the housing. The electrical insulator may comprise a support structure configured to support the electronic unit. The support structure may comprise a cradle configured to secure the electronic unit, at least partly, within the cradle.

[0007] In some embodiments, the electrical insulator comprises a printed circuit board (PCB), and the electronic unit may be fixated to the PCB. This is particularly advantageous, as it reduces the amount of componentry required during manufacture of the battery assembly. The electronic unit itself would, typically, be fixated to a PCB. For example, the electronic unit may comprise a microprocessor fixated to a PCB, and this PCB may also provide the functionality of the electrical insulator, thus obviating the need for an additional insulator. Furthermore, in certain embodiments, the electrical contacts required to electrically couple the electronic unit to respectively the housing and the second electrical terminal may be comprised in the PCB, further simplifying manufacture of the battery assembly, and providing a more convenient retrofittable solution.

[0008] In some embodiments, a surface of the housing may comprise a support structure configured to support the electronic unit. For example, the support structure may be configured on a face of the housing from which the second battery cell terminal protrudes.

[0009] In some embodiments, the support structure may be configured on an internal surface of the housing, and the electronic unit may be mounted on the support structure. The electronic unit may be coated with a protective film to avoid any interference with the electro-chemistry within the battery cell. Additionally, in some embodiments, the housing may comprise a window allowing Radio Frequency (RF) transmissions to pass through the housing via the window. This may be useful where the electronic unit comprises a wireless transmitter or receiver, in which case the window enables the passage of Radio Frequency (RF) signals to and/or from the electronic unit.

[0010] In accordance with some embodiments, the electronic unit may be comprised on a laminated wrap adhered to at least a portion of an external surface of the housing. For example, the electronic unit may be provided in a central portion of the laminated wrap, with the central portion aligned with the end face of the housing comprising the second battery cell terminal. This may simplify manufacturing of the battery assembly.

[0011] The electronic unit may comprise a wireless transmitter configured to wirelessly transmit the measured property of the battery cell. For example, the measured property may be wirelessly transmitted to a battery management system (BMS) located remotely from the battery assembly. Received information may then be used by the BMS to manage operation and usage of the battery assembly.

[0012] In accordance with some embodiments, the electric battery assembly may comprise a current interrupt device (CID). The CID may comprise a pressure detector configured to detect when an internal gas pressure of the battery cell is greater than or equal to a predetermined gas pressure. The CID may be configured to establish an electrical contact with the electronic unit in dependence on the internal gas pressure of the battery cell being greater than or equal to the predetermined gas pressure. In this way, the electronic unit may be used to monitor the state of the CID. The electronic unit may further be configured to transmit a signal to a BMS when the CID is in electrical contact with the electronic unit.

[0013] In accordance with another aspect of the disclosure, there is provided an electric battery, or battery pack, comprising a plurality of the aforementioned electric battery assemblies.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] In the succeeding sections of the disclosure, specific embodiments of the disclosure will be described, by way of non-limiting example only, with reference to the accompanying figures, in which:

[0015] FIG. 1 is a perspective view of an electric battery assembly comprising an electronic unit electrically connected to the battery cell housing and to the second battery cell terminal, which is electrically insulated from the cell housing, in accordance with an embodiment;

[0016] FIG. 2 is a magnified view of the second battery cell terminal and electrically connected electronic unit of FIG. 1;

[0017] FIG. 3 is a cross-sectional side view of the second battery terminal and electrically connected electronic unit of FIGS. 1 and 2;

[0018] FIG. 4 is a cross-sectional side view of the second battery terminal and electrically connected electronic unit, in accordance with an embodiment in which the electrical insulator comprises a printed circuit board (PCB);

[0019] FIG. 5 is a perspective, exploded view of an electric battery assembly comprising a current interrupt device (CID), in accordance with an embodiment;

[0020] FIG. 6 is a perspective view of an electric battery assembly, wherein the electrical insulator comprises a support structure configured to support the electronic unit;

[0021] FIG. 7 is an exploded perspective view of the electric battery assembly of FIG. 6, in which pin contacts are used to establish an electrical connection between the electronic unit and respectively the battery cell housing and the second electrical terminal of the battery cell;

[0022] FIG. 8 is a cross-sectional perspective view of the electric battery assembly of FIG. 6, illustrating the pin contacts;

[0023] FIG. 9 is a perspective view of an electric battery assembly comprising an electronic unit electrically connected to a substrate residing on a battery housing wall, in accordance with an embodiment;

[0024] FIG. 10 is a perspective view of an electric battery assembly comprising an electronic unit mounted inside a battery housing 104 and a window, exposing a portion of the electronic unit, in accordance with an embodiment;

[0025] FIG. 11 is a cross-sectional perspective view of the electric battery assembly of FIG. 10;

[0026] FIG. 12 is a perspective view of an electric battery cell comprising a laminated wrap adhered around a battery cell, including an electronic circuit, in accordance with an embodiment;

[0027] FIG. 13 is a magnified view of the central portion of the laminated wrap comprised in the battery assembly of FIG. 12;

[0028] FIG. 14 is an electric battery assembly comprising a battery cell having a rectangular prismatic form-factor, in which the first and second battery cell terminals are located on opposing faces at opposite ends of the battery cell, in accordance with an embodiment; and

[0029] FIG. 15 is a perspective view of one end face of the electric battery assembly of FIG. 14.

DETAILED DESCRIPTION OF EMBODIMENTS

[0030] In the following illustrative description of embodiments, like numbered reference numerals appearing in different figures will be used to refer to shared features.

[0031] FIG. 1 illustrates a perspective view of electric battery assembly 100, in accordance with an embodiment of the disclosure. Electric battery assembly 100 comprises battery cell 102, having housing 104. Battery cell 102 comprises first 106 and second 108 electrical terminals. First 106 and second 108 electrical terminals may relate respectively to positive and negative battery cell terminals, or to negative and positive battery cell terminals. In other words, either first electrical terminal 106 may relate to a positive battery cell terminal and second electrical terminal 108 to the

negative cell terminal, or vice versa. Either configuration is suitable for use with the herein disclosed embodiments. First **106** and second **108** electrical terminals are connected to electrodes within battery cell **102**. The electrodes relate to the anode and cathode. In addition, the following features are located within battery cell **102**: a separator, electrolyte and current collectors. In combination with the anode and cathode, these features are sometimes referred to as an electrode stack. Battery cell **102** may comprise one or more electrode stacks within it (not shown). First **106** and second **108** electrical terminals enable electric power to be drawn from battery cell **102** by virtue of being connected to one or more electrode stacks within battery cell **102**.

[0032] First electrical terminal **106** is electrically coupled to housing **104**. In this way, housing **104** effectively acts as an electrical terminal for battery cell **102**. Second electrical terminal **108** is electrically insulated from housing **104** by electrical insulator **110**. Electric battery assembly **100** comprises electronic unit **112**. Electronic unit **112** comprises a measurement device, and in some embodiments may also comprise a wireless transmitter for communicating measurement data with, for example, an external remote battery management system (BMS), or with any other device. Electronic unit **112** is electrically connected to second electrical terminal **108**, and to housing **104**. Electrical connection of electronic unit **112** to housing **104** electrically couples electronic unit **112** to first battery cell terminal **106**, since housing **104** is itself electrically coupled to first battery cell terminal **106**. In this way electrical coupling between both first **106** and second **108** electrical battery cell terminals and electronic unit **112** may be established, without requiring a direct physical connection between electronic unit **112** and both electrical battery cell terminals.

[0033] Electrical coupling of first **106** and second **108** battery cell terminals with electronic unit **112** enables the measurement device to measure a property of battery cell **102**. For example, such measurement may comprise, but is not limited to, any one or more of: a current measurement, a voltage measurement, and/or temperature. Additionally, in certain embodiments, electrical coupling of first **106** and second **108** battery cell terminals with electronic unit **112** may enable electric power to be drawn from battery cell **102** to power electronic unit **112**, and the associated measurement device. This is particularly useful in embodiments where electronic unit **112** is not otherwise provided with a separate power source.

[0034] FIG. 2 is a magnified view of a portion of battery assembly **100** of FIG. 1, and illustrates a non-limiting example of how electronic unit **112** may be electrically connected to second battery cell terminal **108** and housing **104**, in accordance with certain embodiments. As illustrated, electronic unit **112** comprises first **114** and second **116** electrical contacts, connecting electronic unit **112** to respectively housing **104** and second electrical terminal **108**. In accordance with some non-limiting embodiments, first **114** and second **116** electrical contacts may be established by any one or more of: welding, wire bonding, conductive adhesion, crimping, soldering, a press-fit contact, and/or sprung contacts. Electronic unit **112** may be directly fastened to electrical insulator **110** by, for example, an adhesive.

[0035] FIG. 3 is a cross-sectional side view of the magnified portion of battery assembly **100** illustrated in FIG. 2. FIG. 3 illustrates electrical insulator **110** providing a support structure for electronic unit **112**. In the illustrated embodiment, electrical insulator **110** forms a snap-fit connection around second electrical terminal **108**, thereby insulating it from housing **104**. In some embodiments, electrical insulator **110** may form an interference fit around second electrical terminal **108**. Electrical insulator **110** may be formed of any electrically insulating material. For example, in some embodiments, electrical insulator **110** may be formed of an electrically insulating plastic material. The precise material of electrical insulator **110** is immaterial for present purposes, provided that it is electrically insulating. Purely for non-limiting illustrative purposes, in FIG. 3, the orientation of first electrical contact **114** has been changed relative to its orientation in FIG. 2, to show the electrical connection of electronic unit **112** with housing **104**. FIG. 3 also clearly shows electrode **118** coupled to second battery cell terminal **108**.

[0036] In some embodiments, the snap-fit or interference fit connection of electrical insulator **110**

to second electrical terminal **108** enables the combined electrical insulator **110** and attached electronic unit **112** to be retrofittable to battery cell **102**, after assembly of battery cell **102**. Such connection also facilitates removal of electrical insulator **110** and electronic unit **112** for battery maintenance.

[0037] In accordance with some embodiments, electrical insulator **110** may comprise a printed circuit board (PCB) to which electronic unit **112** is electrically connected. FIG. **4** is a cross-sectional side view of second electrical terminal **108**, in such an embodiment. Second electrical terminal **108** may comprise terminal cap **108A** and stud **108B**. A physical connection with either cap **108A** or stud **108B** suffices to provide an electrical connection with second electrical terminal **108**. As illustrated, electrical insulator **110** comprises a PCB to which electronic unit **112** is attached. The PCB may be made of a plastic material, such as FR-4. PCB **110** may be configured with cut-out **121** (illustrated in FIG. **5**) complementary in shape to the cross-sectional shape of stud **108B**, and configured to receive stud **108B**. This enables PCB **110** to fit around stud **108B**. Terminal cap **108A** may be configured to hold PCB **110** in place, as illustrated in FIG. **5**, by engaging the end of stud **108B** protruding from cut-out **121** in PCB **110**. PCB **110** may be configured with electrical wiring within its laminate to establish first **114** and second **116** electrical contacts between housing **104**, electronic unit **112** and second electrical terminal **108**. For example, PCB **110** may comprise first electrical contact **114** on its bottom surface, positioned to enable an electrical connection to be established with housing **104**, when PCB **110** is in contact with housing **104**. Similarly, PCB **110** may be configured with second electrical contact **116** positioned at the face of cut-out **121**, enabling an electrical connection to be established with second electrical terminal **108**, via stud **108B**, when PCB **110** is fit around stud **108B**.

[0038] In accordance with some embodiments, electric battery assembly **100** may comprise a current interrupt device (CID). Conventionally, a CID is configured to open-circuit a battery cell, when internal gas pressure is equal to or greater than a predetermined pressure. The predetermined gas pressure may relate to a threshold pressure value within the battery cell, at which continued operation of the battery cell is considered dangerous, and could potentially result in catastrophic battery failure. The CID serves as a safety device to prevent excessive gas pressure build-up within the battery cell, to reduce the likelihood of catastrophic battery failure.

[0039] FIG. **5** is an exploded, perspective view of battery assembly **100** comprising CID **120**, in accordance with an embodiment. CID comprises a pressure detector **122** configured to detect when an internal gas pressure of battery cell **102** is greater than or equal to a predetermined gas pressure. CID **120** may be configured to establish an electrical contact with electronic unit **112** in dependence on the internal gas pressure of battery cell **102** being greater than or equal to the predetermined gas pressure. Electronic unit **112** may be configured to transmit a signal to, for example, an external, remotely located battery management system (BMS) when CID **120** is in electrical contact with electronic unit **112**. The electronic signal being indicative that the internal gas pressure within battery cell **102** is equal to or greater than the predetermined internal gas pressure, and therefore indicative that battery cell **102** may no longer be in a safe operating state. In embodiments where electronic unit **112** comprises a wireless transmitter, the electronic signal may be transmitted to the BMS by the wireless transmitter. Upon receipt of the electronic signal, the BMS may take any necessary action to mitigate for the safety risk associated with the detected build-up of internal gas pressure within battery cell **102**.

[0040] In some embodiments, CID **120** may operate in its known conventional manner. For example, CID **120** may be connected to a fused circuit (not shown in FIG. **5**). When pressure detector **122** detects that the internal gas pressure is equal to or greater than the predetermined gas pressure, CID **120** causes the fuse to blow, thereby open-circuiting battery cell **102**.

[0041] In the embodiment illustrated in FIG. **5**, pressure detector **122** may comprise a diaphragm configured to deform and establish the electrical contact with electronic unit **112**. Additionally, as illustrated in FIG. **5**, electrical insulator **110** may be a PCB. PCB **110** may be provided with an

electrical contact on its bottom face, such that when the diaphragm deforms and overturns, the electrical contact with electronic unit **112** is established.

[0042] FIG. **6** illustrates a further embodiment, in which electrical insulator **110** defines a support structure for receiving electronic unit **112**. For example, electrical insulator **110** may comprise cradle **124** configured to receive electronic unit **112**, at least partly, within it. Electronic unit **112** may be fixated to cradle **124** by any suitable means or device. For example, by any one or more of: an adhesive strip, an interference fit, and/or a snap-fit.

[0043] FIG. **7** is an exploded, perspective view of battery assembly **100** of FIG. **6**. Cradle **124** is clearly illustrated. Electronic unit **112** is fastened to cradle **124** via pin contacts **128**, which also double as electrical contacts, thereby establishing the electrical connection to second electrical terminal **108** and to housing **104**. As illustrated, cradle **124** defines a recess for receiving electronic unit **112**.

[0044] FIG. **8** is a cross-sectional perspective view of assembly **100** of FIG. **7**. The cross-section is taken along second electrical terminal **108** in a plane parallel to the width of battery cell **102**. First pin contact **128A** establishes electrical contact with housing **104**, whilst second pin contact **128B** establishes electrical contact with second electrical terminal **108**. In accordance with some embodiments, pin contacts **128A** and **128B** may comprise spring-loaded pins, configured such that when electronic unit **112** is placed in cradle **124**, the springs are compressed. Compression of the springs displaces first **128A** and second **128B** pin contacts such that electrical contact between electronic unit **112** and respectively housing **104** and second electrical terminal **108** is established.

[0045] FIG. **9** illustrates another embodiment, in which a face of housing **104** comprises a support structure configured to support electronic unit **112**. For example, as shown in FIG. **9**, the support structure may be configured on the top face of housing **104**. This ensures the support structure is provided on the same face from which second terminal **108** protrudes. The support structure may comprise substrate **126**, such as a PCB to which electronic unit **112** is electrically connected. The PCB may be made of plastic material, such as FR-4 or metal (such as aluminum or copper), and the metal used to define the PCB traces may be similar to, or different from the metal used in housing **104**. In other words, the material of housing **104** does not restrict the choice of material to define the PCB traces. FIG. **9** further illustrates a non-limiting example of how electronic unit **112** may be electronically connected to second battery cell terminal **108** and housing **104**. As illustrated, first **114** and second **116** electrical contact may be provided to connect electronic unit **112** to respectively housing **104** and second electrical terminal **108** via substrate **126**.

[0046] FIG. **10** illustrates a further embodiment, in which electronic unit **112** is mounted inside, on an internal surface of housing **104**. Electrical connection with first **106** and second **108** battery cell terminals may occur in the same way as previously described. Electronic unit **112** may be coated to avoid any interference with the electro-chemistry within battery cell **102**, i.e. to ensure there is no contamination of or undesired chemical reactions with the electrochemistry within battery cell **102** (e.g., by using a protective film, coating or conformal coating). Electronic unit **112** may be attached to an inner surface of housing **104** by any suitable means, such as an adhesive.

[0047] In some embodiments, where electronic unit **112** may comprise wireless transmitter, such as a Radio Frequency (RF) transmitter, housing **104** may comprise window **130**, configured to enable RF transmissions through housing **104**. For example, and as illustrated in FIG. **10**, window **130** may enable RF coupling between electronic unit **112** and RF transmission line **132**. For non-limiting illustrative purposes only, only a portion of RF transmission line **132** is illustrated. RF transmission may be between an antenna internal to the battery cell **102** and an antenna external to the battery cell **102**, such as transmission line **132**, through housing **104** via window **130**. Window **130** may be made of any type of material permitting the transmission of RF signals. In other words, any material that presents a low loss to the transmission of RF energy. For example, different types of plastics, glasses, ceramics or other RF transparent material may be used for window **130**.

[0048] FIG. **11** is a cross-sectional perspective view of battery, taken along a length of battery cell

102 of FIG. **10**. FIG. **11** clearly shows electronic unit **112** affixed to an internal surface of housing **104**, and positioned below window **130** to enable the transmission of RF signals between electronic unit **112** and RF transmission line **132**.

[0049] FIG. **12** is a perspective view of another embodiment, in which electronic unit **112** is comprised in laminated wrap **140**. Laminated wrap **140** is adhered to battery cell **102**. As in previously described embodiments, electronic unit **112** may be comprised on a flexible PCB. Laminated wrap **140** may comprise a thermally conductive sheet which interfaces with one or more surfaces of housing **104**, thereby providing a thermal interface for cooling or warming. Laminated wrap **140** may comprise an electrically insulating sheet, thereby insulating battery cell **102** from neighboring cells. Laminate wrap **140** may wrap around at least a portion the external surfaces of housing **104**, or the entire circumference of housing **104** as shown in FIG. **12**. Laminated wrap **140** may be constructed of any flexible material having the desired thermal and adhesive characteristics. For example, any kind of plastic or polymer such as acrylic, vinyl, polyurethane, polyester, or PVC. Yet a further advantage of using a laminated wrap, is that depending on the selected material, laminated wrap **140** may as act as a fire retardant, for improved safety.

[0050] FIG. **13** is a plan view of unwrapped laminated wrap **140** of FIG. **12**. In accordance with the illustrated embodiment, electronic unit **112** is provided in a central portion of laminated wrap **140**, on the portion of laminated wrap **140** which aligns with the end face of housing **104** comprising second battery cell terminal **108**. First **114** (not shown) and second **116** electrical contacts configured to connect electronic unit **112** to respectively housing **104** and second electrical terminal **108** may be comprised in laminate wrap **140**. For example, first electrical contact **114** may be located under electronic unit **112**, passing through laminated wrap **140**, thereby establishing an electrical connection between the electronic unit **112** and the housing **104**; second electrical contact **116** may be embedded within laminate wrap **140** structure, and may comprise ring-like protrusion **146** provided to establish an electrical connection to second battery cell terminal **108**, as illustrated in FIG. **12**. One of the benefits of the embodiment shown in FIGS. **12-13** is that it does not require additional design of battery cell **102** when it leaves factory. In other words, the laminated structure described herein is retrofittable and may be suitable for a wide range of battery cell types.

[0051] The means and embodiments disclosed herein for electrically coupling an electronic unit to the electrical terminals of a battery cell are particularly convenient for use with battery cells having a form-factor where the distance of separation between the electrical terminals is large, and/or where the electrical terminals are located on different faces of the battery cell. As mentioned previously, electrically coupling one of the battery terminals to the battery housing, effectively results in the housing doubling as one of the battery terminals. Therefore, all that is required to establish a circuit between the battery terminals and an electronic unit, is an electrical connection between the electronic unit and the electrical terminal of the cell that is electrically insulated from the housing, and an electrical connection between the electronic unit and the housing. This may significantly reduce the amount of electrical wiring required to electrically couple the electronic unit to the battery cell.

[0052] FIGS. **14** and **15** are examples of battery cells having different form-factors, where use of the herein disclosed means for electrically coupling electronic unit **112** to the battery cells may be particularly advantageous. FIG. **14** illustrates a rectangular-shaped prismatic cell, having a length greater than its width, in which the first **106** and second **108** electrical terminals are located on opposing faces of battery cell **102**. More specifically, the first **106** and second **108** electrical terminals are located on opposing end faces of battery cell **102**, the end faces being located at opposite ends of the cell's length. Coupling electronic unit **112** to housing **104** and second electrical terminal **108** as previously disclosed, facilitates assembly of battery assembly **100**, for reasons previously disclosed. FIG. **15** is a perspective view of the end face of battery cell **102** comprising second electrical terminal **108** and electronic unit **112**, illustrated in FIG. **14**.

[0053] It is to be appreciated that in accordance with embodiments of the disclosure, the electronic

unit may be positioned at any location on the housing, as this suffices to electrically couple the electronic unit to the first electrical terminal. It is simply for convenience that the enclosed illustrated embodiments depict the electronic unit being located on the battery cell face comprising the second electrical terminal, as this minimizes the length of electrical contact required to electrically couple the electronic unit to the second electrical terminal.

[0054] It is also to be appreciated that first **106** and second **108** electrical terminals may relate respectively to the positive and negative battery cell terminals, or respectively to the negative and positive battery cell terminals. In other words, it is immaterial which electrical terminal housing **104** is electrically coupled to, and which electrical terminal is electrically insulated from housing **104**.

[0055] The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, and are not intended to be exhaustive or limiting to the scope of the disclosure. Many modifications and variations of the disclosed embodiments will be apparent to those of ordinary skill in the art from the herein disclosed embodiments, without departing from the scope of the disclosure. The terminology used herein to disclose the embodiments of the disclosure was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

[0056] It is appreciated that certain features of the disclosure, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the disclosure, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination or as suitable in any other described embodiment of the disclosure. Certain features described in the context of various embodiments are not to be considered essential features of those embodiments, unless the embodiment is inoperative without those elements.

[0057] Although the disclosure has been described in conjunction with specific embodiments thereof, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the spirit and broad scope of the appended claims.

Claims

1. An electric battery assembly comprising: a battery cell having an exterior housing and first and second electrical terminals enabling power to be drawn from the battery cell, the housing being electrically coupled to the first electrical terminal and electrically insulated from the second electrical terminal; an electronic unit comprising a measurement device; and wherein the electronic unit is electrically connected to the second electrical terminal and to the housing, thereby electrically coupling the electronic unit to the first battery cell terminal via the housing, enabling the measurement device to measure a property of the battery cell.
2. The electric battery assembly of claim 1, comprising: an electrical insulator configured to insulate the second electrical terminal from the housing.
3. The electric battery assembly of claim 2, wherein the electrical insulator comprises a support structure configured to support the electronic unit.
4. The electric battery assembly of claim 3, wherein the electrical insulator comprises a cradle configured to secure the electronic unit at least partly within the cradle.
5. The electric battery assembly of claim 3 or 4, wherein the electrical insulator comprises a printed circuit board “PCB”, and the electronic unit is fixated to the PCB.
6. The electric battery assembly of any one of claims 3 to 5, wherein the electrical insulator comprises first and second contacts, the first contact being configured to electrically couple the electronic unit to the housing, and the second contact being configured to electrically couple the

electronic unit to the second electrical terminal.

7. The electric battery assembly of claim 6, wherein the first and second contacts are configured to fasten the electronic unit to the electrical insulator.

8. The electric battery assembly of claim 1 or 2, wherein a surface of the housing comprises a support structure configured to support the electronic unit.

9. The electric battery assembly of claim 8, wherein the support structure is configured on an internal surface of the housing, and the electronic unit is mounted on the support structure.

10. The electric battery of claim 9 wherein the electronic unit is coated with a protective film.

11. The electric battery assembly of claim 9 or 10, wherein the housing comprises a window enabling RF transmissions to pass through the window.

12. The electric battery assembly of claim 1 or 2, wherein the electronic unit is comprised on a laminated wrap adhered to at least a portion of an external surface of the housing.

13. The electric battery assembly of any one of claims 8 to 12, comprising first and second contacts, the first contact being configured to electrically couple the electronic unit to the housing, and the second contact being configured to electrically couple the electronic unit to the second electrical terminal.

14. The electric battery assembly of any preceding claim, wherein the electronic unit is configured to draw electrical power from the battery cell.

15. The electric battery assembly of any preceding claim, wherein the electronic unit comprises a wireless transmitter configured to wirelessly transmit the measured property of the battery cell.

16. The electric battery assembly of claim 14 or 15, comprising a current interrupt device "CID", the CID comprising a pressure detector configured to detect when an internal gas pressure of the battery cell is greater than or equal to a predetermined gas pressure, the CID being configured to establish an electrical contact with the electronic unit in dependence on the internal gas pressure of the battery cell being greater than or equal to the predetermined gas pressure; and wherein the electronic unit is configured to transmit a signal to a battery management system when the CID is in electrical contact with the electronic unit.

17. The electric battery assembly of claim 16, wherein the pressure detector comprises a diaphragm configured to deform and establish the electrical contact with the electronic unit.

18. The electric battery assembly of any preceding claim, wherein the first and second electrical terminals are respectively the positive and negative electrical terminals of the battery cell, or respectively the negative and positive electrical terminals of the battery cell.

19. The electric battery assembly of any preceding claim, wherein the battery cell is a prismatic battery cell.

20. The electric battery assembly of claim 19, wherein the first and second electrical terminals are located at different faces of the prismatic cell.

21. The electric battery assembly of claim 20, wherein the first and second electrical terminals are located on opposing faces of the prismatic cell.

22. An electric battery comprising a plurality of electric battery assemblies of any one of claims 1 to 21.
